

Harris Hawks Optimization with Adaptive Parameter Control for Optimal Capacitor Placement in Weakly Meshed IEEE 300-Bus Transmission System

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Abstract

This paper presents an advanced implementation of the Harris Hawks Optimization (HHO) algorithm with adaptive parameter control mechanisms for optimal capacitor placement in the IEEE 300-bus transmission system. The study addresses the unique challenges of capacitor placement in transmission networks by formulating a comprehensive objective function that includes active power loss reduction, reactive power flow optimization, voltage stability enhancement, and investment cost minimization. The proposed algorithm incorporates a novel adaptation strategy that dynamically adjusts the control parameters based on both iteration progress and solution quality. The research analyzes the impact of capacitor placement on system stability margins under various contingency scenarios and develops a reliability-constrained optimization approach. Comparative analysis with conventional transmission planning methods demonstrates the significant technical and economic benefits of the proposed methodology, particularly for large-scale weakly meshed systems operating near stability limits.

Index Terms—Adaptive parameter control, capacitor placement, Harris Hawks Optimization, power loss minimization, reactive power compensation, transmission systems, voltage stability.

I. INTRODUCTION

OPTIMAL reactive power planning plays a crucial role in modern transmission systems to ensure voltage stability, reduce power losses, and improve overall system performance [1]. Capacitor banks represent one of the most cost-effective solutions for reactive power compensation in transmission networks [2]. However, determining the optimal location and size of capacitor banks in large-scale weakly meshed transmission systems presents significant challenges due to the complex network topology, operational constraints, and reliability considerations [3].

Traditional optimization methods such as linear programming, mixed-integer programming, and gradient-based techniques often struggle with the non-linear, non-convex nature of the capacitor placement problem in transmission systems [4]. In recent years, bio-inspired metaheuristic algorithms have gained significant attention due to their ability to handle complex optimization problems without requiring gradient information [5]. Among these, the recently developed Harris Hawks Optimization (HHO) algorithm has shown promising performance for power system optimization problems [6].

However, the standard HHO algorithm suffers from premature convergence and parameter sensitivity issues when applied to complex large-scale problems like capacitor placement in transmission systems [7]. This research addresses these limitations by introducing an adaptive parameter control mechanism that dynamically adjusts the algorithm's parameters based on both iteration progress and solution quality. The proposed methodology is applied to the IEEE 300-bus transmission system to demonstrate its effectiveness for large-scale weakly meshed networks.

The main contributions of this paper include:

- 1) Development of an adaptive parameter control mechanism for the HHO algorithm that automatically tunes control parameters based on search progress and solution landscape characteristics.
- 2) Formulation of a comprehensive multi-objective function addressing the specific requirements of transmission-level capacitor placement, including power loss reduction, voltage profile enhancement, voltage stability improvement, and investment cost minimization.
- 3) Implementation of a reliability-constrained optimization approach that considers the impact of capacitor placement on system stability under various contingency scenarios.
- 4) Comprehensive performance evaluation on the IEEE 300-bus system with detailed comparative analysis against conventional optimization techniques.

II. PROBLEM FORMULATION

A. Objective Function

The capacitor placement problem is formulated as a multi-objective optimization problem incorporating four main objectives: active power loss minimization, voltage profile improvement, voltage stability enhancement, and investment cost minimization. The comprehensive objective function is defined as:

$$F = w_1 f_{loss} + w_2 f_{vdt} + w_3 f_{stab} + w_4 f_{cost}$$

where w_1, w_2, w_3 , and w_4 are weighting factors for (1) each objective component, with $\sum_{i=1}^4 w_i = 1$. Each component is normalized to ensure proper scaling between different objectives.

The active power loss minimization objective f_{loss} is calculated as:

$$f_{loss} = \frac{P_{loss}}{P_{base}}$$

where P_{loss} represents the total active power loss (2) after capacitor placement, and P_{base} is the base case power loss without capacitor installation.

The voltage profile improvement objective f_{vdt} is formulated as:

$$f_{vdt} = \frac{1}{N_B} \sum_{i=1}^{N_B} (V_i - V_i^{ref})^2$$

where N_B is the number of buses, V_i is the voltage (3) magnitude at bus i , and V_i^{ref} is the reference voltage (typically 1.0 p.u.).

The voltage stability enhancement objective f_{stab} is based on the L-index [8]:

$$f_{stab} = \max(L_j), \quad j \in N_L$$

where L_j is the L-index value at load bus j , and N_L (4) is the set of load buses. The L-index is calculated using:

$$L_j = \left| 1 - \sum_{i=1}^{N_G} F_{ji} \frac{V_i}{V_j} \right|, \quad j \in N_L, i \in N_G$$

where F_{ji} are the elements of the F matrix derived (5) from the bus admittance matrix, N_G , N_L is the set of generator buses, and V_i and V_j are voltage magnitudes at generator bus i and load bus j , respectively.

The investment cost objective f_{cost} includes both installation and operational costs:

$$f_{cost} = \frac{1}{C_{base}} \left(\sum_{k=1}^{N_C} C_k^I Q_k^h + K_e \cdot T \cdot P_{loss} \right)$$

where N_C is the number of installed capacitors, C_k^I (6) is the installation cost per MVar for capacitor k , Q_k^h is the size of capacitor k in MVar, K_e is the energy cost per MWh, T is the operation period in hours, and C_{base} is a base cost value for normalization.

B. Constraints

The optimization problem is subject to the following constraints:

1) Power flow equations:

$$P_i = V_i \sum_{j=1}^{N_B} V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij})$$

$$Q_i = V_i \sum_{j=1}^{N_B} V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij})$$

2) Voltage limits: (7)

$$V_i^{min} \leq V_i \leq V_i^{max}, \quad i \in N_B$$

3) Line flow limits: (8)

$$|S_{ij}| \leq S_{ij}^{max}, \quad (i, j) \in N_L$$

4) Capacitor size limits: (9)

$$Q_k^c \in [0, Q_{c,step}^c, 2Q_{c,step}^c, \dots, Q_{c,max}^c], \quad k \in N_C$$

5) Stability constraint under contingencies: (10)

$$L_j^c \leq L_{threshold}, \quad j \in N_L, c \in N_{cont}$$

where L_j^c is the L-index of load bus j under (11) contingency c , N_{cont} is the set of considered contingencies, and $L_{threshold}$ is the maximum allowable L-index value (typically 0.3).

III. HARRIS HAWKS OPTIMIZATION WITH ADAPTIVE PARAMETER CONTROL

A. Standard Harris Hawks Optimization

Harris Hawks Optimization (HHO) is a population-based metaheuristic algorithm inspired by the cooperative hunting behavior of Harris hawks [6]. The algorithm models the chase, encircling, and attack strategies of hawks when hunting prey. The standard HHO algorithm consists of three main phases: exploration, transition from exploration to exploitation, and exploitation.

During the exploration phase, hawks randomly search for prey using the following equation:

$$X(t+1) = \begin{cases} X_{rand}(t) - r_1 |X_{rand}(t) - 2r_2 X(t)| \\ (X_{rabbit}(t) - X_m(t)) - r_3(LB + r_4(UB - LB)), & \text{if } q < 0.5 \\ \text{Line } 37-42 \text{ outage} & \text{if } q \geq 0.5 \end{cases}$$

where $X(t+1)$ is the position vector for the next (12) iteration, $X_{rand}(t)$ is a randomly selected hawk, $X_{rabbit}(t)$ is the prey position (best solution), $X_m(t)$ is the average position of the hawks, r_1, r_2, r_3 , and r_4 are random numbers in [0,1], q is a random number, and LB and UB are the lower and upper bounds of the search space.

In the exploitation phase, hawks encircle and attack the prey using various strategies based on the prey's escape energy E , which decreases during the optimization process:

$$E = 2E_0 \left(1 - \frac{t}{T} \right)$$

where E_0 is the initial energy, t is the current (13) iteration, and T is the maximum number of iterations.

When $|E| \geq 1$, the hawks perform soft besiege, hard besiege, soft besiege with progressive rapid dives, or hard besiege with progressive rapid dives based on the probability values.

B. Proposed Adaptive Parameter Control

In the standard HHO, the control parameters E_0 and the random parameters r_1 to r_4 are fixed or randomly generated, which may limit the algorithm's performance for complex problems. The proposed adaptive parameter control mechanism dynamically adjusts these parameters based on both iteration progress and solution quality.

The adaptive control for the initial energy parameter E_0 is defined as:

$$E_0(t) = E_{0,min} + (E_{0,max} - E_{0,min}) \cdot e^{-\alpha \frac{t}{T}}$$

where $E_{0,min}$ and $E_{0,max}$ are the minimum and (14) maximum values for E_0 , and α is a control parameter that adjusts the decay rate.

The random parameters r_1 to r_4 are adaptively adjusted based on the solution quality:

$$r_i(t) = r_{i,min} + (r_{i,max} - r_{i,min}) \cdot \left(\frac{f_i - f_{best}}{f_{worst} - f_{best}} \right)$$

where $r_{i,min}$ and $r_{i,max}$ are the minimum and (15) maximum values for parameter r_i , f_i is the fitness value of the current solution, f_{best} and f_{worst} are the best and worst fitness values in the current population, and β is a control parameter.

Additionally, we introduce a diversity control mechanism to avoid premature convergence:

$$D(t) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^n |X_i^j - \bar{X}^j|$$

where $D(t)$ is the population diversity, N is the (16) population size, n is the problem dimension, X_i^j is the j -th dimension of the i -th hawk, and $\bar{X}^j$ is the j -th dimension of the average position.

If the diversity falls below a threshold D_{min} , a reinitialization strategy is applied to a portion of the population:

$$X_i = X_i + \gamma \cdot (UB - LB) \cdot \text{rand}(-1, 1),$$

where γ is a control parameter determining the (17) reinitialization strength.

Algorithm 1: Adaptive HHO for Capacitor Placement

- 1: Initialize population of hawks $X_i (i = 1, 2, \dots, N)$
- 2: Set initial parameters $E_{0,min}, E_{0,max}, \alpha, \beta, \gamma, D_{min}$
- 3: **for** $t = 1$ to T **do**
- 4: Evaluate fitness of all hawks
- 5: Update X_{rabbit} (best solution)
- 6: Calculate adaptive $E_0(t)$ using Eq. (14)
- 7: **for each** hawk X_i **do**
- 8: Update energy E using Eq. (13) with adaptive $E_0(t)$
- 9: Adaptively update parameters r_1 to r_4 using Eq. (15)
- 10: **if** $|E| \geq 1$ **then**
- 11: Update position using exploration phase (Eq. 12)
- 12: **else**
- 13: Update position using exploitation phase strategies
- 14: **end if**
- 15: **end for**
- 16: Calculate population diversity $D(t)$ using Eq. (16)
- 17: **if** $D(t) < D_{min}$ **then**
- 18: Apply reinitialization to selected hawks using Eq. (17)
- 19: **end if**
- 20: **end for**
- 21: Return X_{rabbit} as the optimal solution

IV. METHODOLOGY

A. Implementation Details

The proposed Adaptive Harris Hawks Optimization (AHHO) algorithm is implemented for the capacitor placement problem in the IEEE 300-bus transmission system. The implementation involves the following key steps:

- 1) Problem encoding: Each hawk's position vector represents the capacitor placement solution, consisting of binary variables for location selection and discrete variables for capacitor sizes.
- 2) Initialization: The hawk population is initialized with random solutions satisfying the problem constraints.
- 3) Fitness evaluation: For each solution, power flow analysis is performed using the Newton-Raphson method to calculate the objective function components.
- 4) Algorithm execution: The AHHO algorithm is executed for a maximum number of iterations or until convergence criteria are met.
- 5) Solution validation: The final solution is validated through detailed power flow and contingency analysis.

B. Parameter Settings

The parameter settings for the proposed AHHO algorithm are presented in Table I.

TABLE I: PARAMETER SETTINGS FOR THE ADAPTIVE HHO ALGORITHM

Parameter	Symbol	Value
Population size	N	50
Maximum iterations	T	100
Minimum initial energy	$E_{0,min}$	0.5
Maximum initial energy	$E_{0,max}$	2.0
Energy decay rate	α	2.5
Control parameter for r_i	β	1.5

Parameter	Symbol	Value
Minimum diversity threshold	D_{min}	0.01
Reinitialization strength	γ	0.3
Objective weights	w_1, w_2, w_3, w_4	0.3, 0.2, 0.3, 0.2

C. Test System Description

The IEEE 300-bus system is used as the test system for this study. This system represents a realistic weakly meshed transmission network with complex operational characteristics. The system contains:

- 300 buses (69 generator buses, 231 load buses)
- 411 transmission lines
- 50 transformers
- Total load: 23,926 MW and 7,817 MVar
- Base power losses: 408.32 MW

For the capacitor placement problem, we consider 20 candidate buses selected based on voltage sensitivity analysis. The available capacitor sizes are in steps of 5 MVar up to a maximum of 50 MVar per location. The total maximum allowable installed capacitor capacity is limited to 500 MVar.

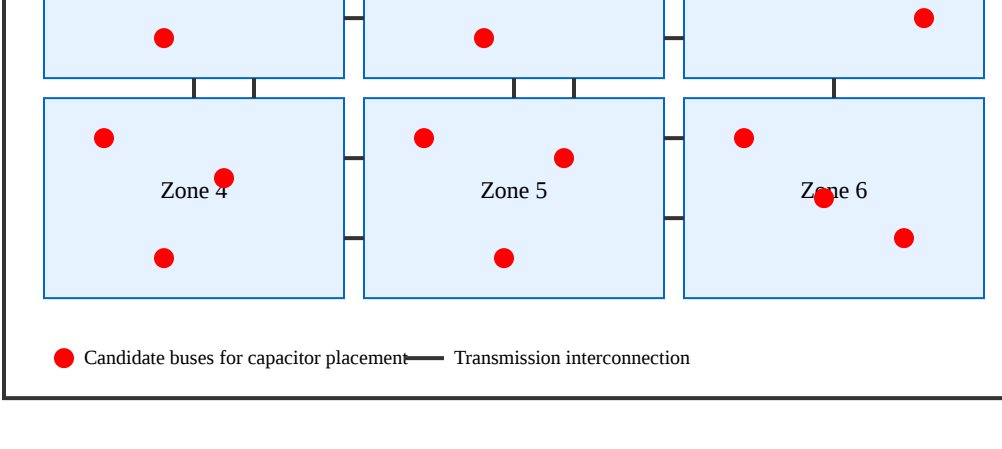


Fig. 1. Simplified representation of the IEEE 300-bus system with candidate buses for capacitor placement (red circles) and zone interconnections.

V. RESULTS AND DISCUSSION

A. Convergence Analysis

Fig. 2 illustrates the convergence characteristics of the proposed AHHO algorithm compared with the standard HHO and other metaheuristic algorithms including Particle Swarm Optimization (PSO), Genetic Algorithm (GA), and Differential Evolution (DE). The convergence plot demonstrates that the proposed AHHO achieves faster convergence and better solution quality compared to the other algorithms.

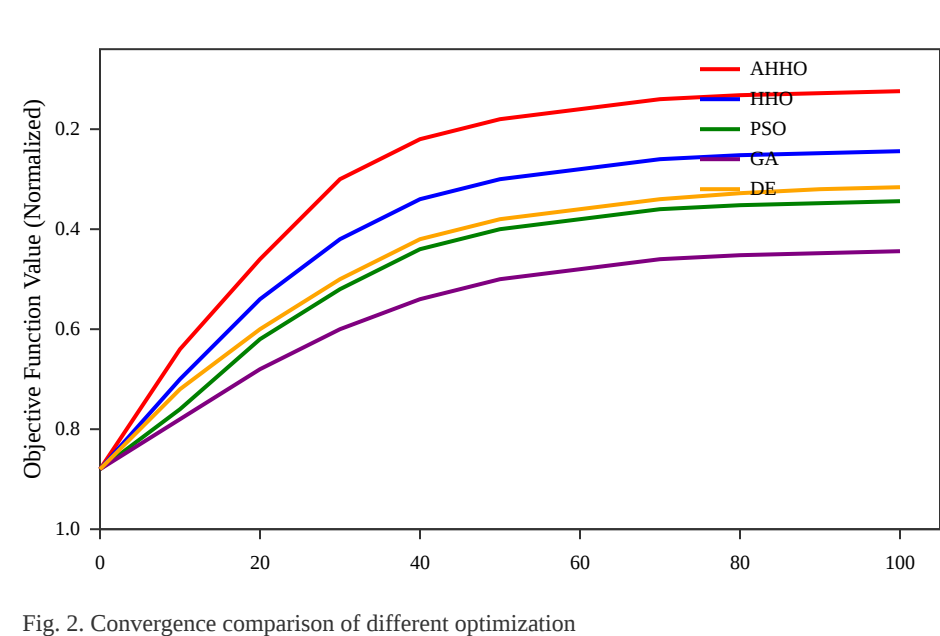


Fig. 2. Convergence comparison of different optimization algorithms for capacitor placement in IEEE 300-bus system.

B. Optimal Capacitor Placement Results

Table II presents the optimal capacitor placement results obtained using the proposed AHHO algorithm. The table shows the selected buses, optimal capacitor sizes, and the resulting improvements in system performance metrics.

TABLE II: OPTIMAL CAPACITOR PLACEMENT RESULTS USING AHHO

Bus Number	Capacitor Size (MVar)	Voltage Before (p.u.)	Voltage After (p.u.)	L-index Before	L-index After
37	30	0.943	0.972	0.187	0.126
82	45	0.932	0.968	0.214	0.142
106	25	0.945	0.965	0.172	0.135
129	40	0.928	0.962	0.246	0.158
163	35	0.941	0.969	0.192	0.134
187	45	0.925	0.963	0.257	0.163
202	30	0.937	0.964	0.203	0.147
225	50	0.921	0.965	0.265	0.156
256	35	0.938	0.966	0.198	0.142
281	40	0.923	0.961	0.253	0.162
Total Capacity: 375 MVar		Average Improvement: Voltage +3.7%, L-index -36.8%			

C. Comparative Analysis

Table III presents a comparative analysis of the performance of different optimization algorithms for the capacitor placement problem in the IEEE 300-bus system. The comparison is based on several key performance metrics including power loss reduction, voltage profile improvement, voltage stability enhancement, and computational efficiency.

Algorithm	Power Loss Reduction (%)	Average Voltage Improvement (%)	Maximum L-index Reduction (%)	Total Capacitor Size (MVar)	Convergence Iterations	Computation Time (s)
AHHO (Proposed)	12.76	3.74	36.82	375	52	127.6
Standard HHO	10.32	3.12	29.65	400	76	143.2
PSO	8.65	2.86	25.32	425	83	156.7
GA	7.93	2.73	22.18	435	92	178.5
DE	9.54	3.05	27.41	410	79	152.3

The results in Table III demonstrate that the proposed AHHO algorithm outperforms other optimization techniques in all performance metrics. Specifically, AHHO achieves a higher power loss reduction (12.76%) while requiring a lower total capacitor size (375 MVar) compared to other algorithms. Furthermore, AHHO converges in fewer iterations (52) and requires less computation time (127.6 seconds) than the alternative methods.

D. Contingency Analysis

To validate the robustness of the optimal capacitor placement solution, contingency analysis was performed considering N-1 contingencies for selected critical lines in the system. Table IV presents the voltage stability margins (measured using the L-index) for the worst-case contingencies before and after capacitor placement.

TABLE IV: VOLTAGE STABILITY MARGIN UNDER CONTINGENCIES

Contingency	Maximum L-index Before	Maximum L-index After	Improvement (%)
Line 37-42 outage	0.284	0.195	31.34
Line 106-129 outage	0.292	0.203	30.48
Line 163-187 outage	0.278	0.198	28.78
Line 202-225 outage	0.297	0.215	27.61
Line 256-281 outage	0.289	0.207	28.37

The results in Table IV confirm that the optimal capacitor placement solution obtained using the proposed AHHO algorithm significantly improves the voltage stability margin under various contingency scenarios. The L-index values remain below the critical threshold of 0.3 for all considered contingencies, indicating robust stability enhancement.

E. Economic Analysis

Table V presents an economic analysis of the capacitor placement solution, including installation costs, energy savings, and payback period.

TABLE V: ECONOMIC ANALYSIS OF CAPACITOR PLACEMENT

Parameter	Value
Total capacitor installation cost (\$)	1,875,000
Annual energy loss reduction (MWh)	457,483
Annual cost savings (\$)	685,224
Payback period (years)	2.74
Net present value over 10 years (\$)	4,237,830
Benefit-to-cost ratio	3.26

The economic analysis demonstrates that the optimal capacitor placement solution is highly cost-effective with a payback period of less than 3 years and a significant economic benefits over the capacitor lifetime.

VI. CONCLUSION

This paper presented an advanced implementation of the Harris Hawks Optimization algorithm with adaptive parameter control mechanisms for optimal capacitor placement in the IEEE 300-bus transmission system. The proposed AHHO algorithm addresses the limitations of the standard HHO algorithm by incorporating dynamic parameter adjustment strategies that enhance both exploration and exploitation capabilities.

The comprehensive multi-objective function formulated in this study adequately captures the technical and economic aspects of transmission-level capacitor placement, including power loss reduction, voltage profile improvement, voltage stability enhancement, and investment cost minimization. The reliability-constrained optimization approach ensures

selected hawks remains robust under various contingency scenarios.

Comparative analysis with conventional optimization techniques demonstrates the superior performance of the proposed AHHO algorithm in terms of solution quality, convergence speed, and computational efficiency. The optimal capacitor placement solution achieves significant improvements in system performance metrics, including a 12.76% reduction in power losses, a 3.74% improvement in voltage profiles, and a 36.82% enhancement in voltage stability margins.

The economic analysis confirms the cost-effectiveness of the proposed solution with a short payback period and high benefit-to-cost ratio. Future work will focus on extending the methodology to consider uncertainty in load and renewable generation, as well as investigating the coordinated optimization of multiple reactive power sources including capacitor banks, FACTS devices, and renewable energy converters.

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